

Cardiovascular Disorders

Learning Objectives

- ⇒ To have an overall review of cardiovascular diseases
- ⇒ To know about the clinical features of coronary artery disease (CAD including IHD and MI) and congestive cardiac failure (CCF)
- ⇒ To know about the necessary physical examinations to be performed in the above said diseases
- ⇒ To know about the laboratory investigations to be ordered in above said diseases and interpretations of the investigations

Introduction

General symptoms of cardiac disease should be studied very carefully, as the gravity of the problem of any individual depends on the general condition of the patient. Many of the symptoms manifested due to the cardiovascular dysfunction will show symptoms of other systemic dysfunction, that is the reason the clinician has to be very careful in diagnosing the cardiac disease.

- ✱ In the investigation of the cardiac disease, the various methods have roughly the following relative values: history 40%, physical examination 25%, electrocardiography 20%, radiography 10% and pathological and other special methods of diagnosis 5%. In other words, 65% of the cases of cardiac disease can be accurately diagnosed by eliciting the history and methodical physical examination.
- ✱ Cardinal principle useful in evaluation of suspected heart disease – myocardial/ coronary function may be adequate at rest but inadequate during activity, but reverse mechanism is rare in organic heart disease.
- ✱ Symptoms caused by heart disease result most commonly from
 - ✱ Myocardial ischemia – chest discomfort which can manifest due to mismatch between the supply and demand of oxygen to the myocardium
 - ✱ Disturbance in contraction/ relaxation of myocardium (reduced pumping activity) – weakness, fatiguability associated with cyanosis, hypotension, syncope, increased intravascular pressure behind failing ventricle
 - ✱ Obstruction to blood flow (abnormal fluid accumulation – dyspnoea, orthopnoea, systemic/ pulmonary oedema, CCF)
 - ✱ Arrhythmias (occurs abruptly, disappears rapidly) – palpitation, angina, dyspnoea, hypotension, syncope
 - ✱ Complete diagnosis of cardiac disease:
 - ✱ Underlying etiology – congenital, infectious, hypertensive, ischemic in origin
 - ✱ Anatomic abnormalities – chambers involved? Which valves affected? Is there pericardial involvement? Has there been an MI?
 - ✱ Physiological disturbances – is arrhythmia present? Is there evidence of CCF, IHD?
 - ✱ Extent of functional disability – how strenuous the exercise should be to elicit the symptoms?
 - ✱ The recognition of ischemia is not sufficient to plan therapeutic strategy as there can be AS, coronary atherosclerosis, severe anemia or thyrotoxicosis
 - ✱ As far as symptoms of cardiac disease are concerned, the person may have varying degrees of breathlessness (breathlessness on exertion, paroxysmal nocturnal dyspnoea, orthopnoea, dyspnoea at rest), chest discomfort, cough with pink frothy sputum, pain in the right upper quadrant, bilateral pedal oedema, loss of appetite, nausea or vomiting, fullness of the abdomen; syncope, oliguria and nocturia.
 - ✱ As far as disease of blood vessels are concerned, the person may have pain in the lower limb with intermittent claudication, blackish discolouration

of the lower limb, shininess of the skin, loss of body hairs where reduced blood supply is observed.

As far as physical examination is concerned, it can be further categorized into general and systemic examination. General examination includes assessment of blood pressure; assessment of peripheral arterial pulsation; pedal oedema; assessment of jugular venous pulsation on the right side.

Systemic examination include inspection, palpation and auscultation of chest in the cardiac assessment. Laboratory examinations include CBC, ESR, electrocardiogram (ECG), chest x ray, echocardiogram, tread mill test (TMT), assessment of cardiac enzyme markers and invasive technique such as cardiac catheterization.

In the following pages, we are going to discuss coronary artery disease; ischaemic heart disease and myocardial infarction and congestive cardiac failure.

Coronary Artery Disease, IHD & MI

Coronary arteries are of two in number, right and left which arise from the right and left coronary sinuses in the root of aorta just above the aortic valve orifice. Coronary Artery Disease (see Figure 68) (CAD) involves impairment of blood flow through coronary arteries, most commonly by atheromas.

Clinical presentations include silent ischemia, angina pectoris, acute coronary syndromes (unstable angina and myocardial infarction) and sudden cardiac death. Diagnosis is done by symptoms, ECG, tread mill test (stress testing i.e. TMT) and sometimes by coronary angiogram.

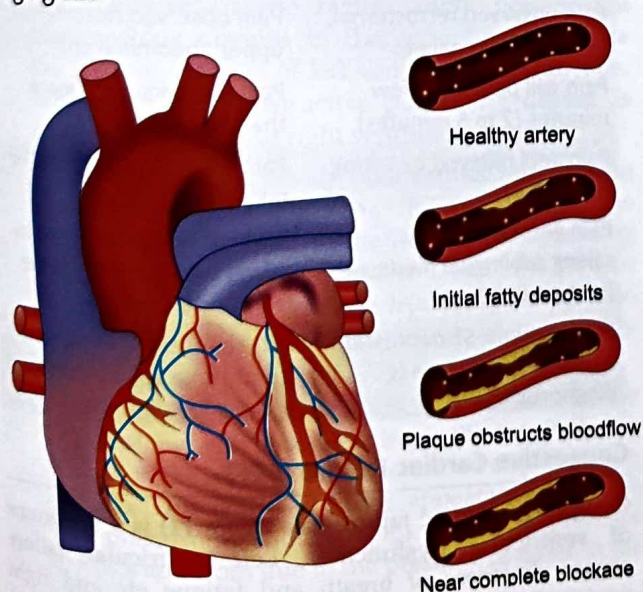


Figure 68: Mechanism of coronary artery disease

Coronary Artery Disease (CAD) is 3rd most common cause of deaths among all causes of death. This is

more commonly seen in males in between the ages of 25 to 34 years; this is commonly seen in women after menopausal age; and after age of 75 years, this is seen equally in both genders. Angina pectoris is a form of CAD.

- CAD is due to the deposition of atheromas in subintimal layer of coronary arteries. Less often, coronary spasm is the reason for CAD in very cold environment. Rare causes of CAD are coronary artery embolism; dissecting aortic aneurysm; conditions which cause tachycardia.
- Coronary atherosclerosis is often irregularly distributed in different vessels, but typically occurs at points of turbulence (vessel bifurcations). As atheromatous plaque grows, the arterial lumen progressively narrows, resulting in ischemia. The degree of ischemia happening in the patient leads to the manifestation of acute coronary syndromes which may vary from trivial stable angina to severe myocardial infarction.
- Coronary artery spasm is a transient, focal increase in vascular tone, markedly narrowing the lumen and reducing the blood flow, typically occurring in cold environment.
- Risk factors for the development of CAD are high blood levels of low density lipoproteins (LDL); diabetes mellitus, smoking, obesity, physical inactivity.
- Diagnosis is done by lipid profile test, coronary angiogram.

Ischaemic Heart Disease & Myocardial Infarction

These two conditions are categorized under acute coronary syndromes where in the patient may complain of chest discomfort in the form of heaviness of the chest or pain which may radiate from retrosternal region to left shoulder, sometimes even upto the left finger.

The pain may also radiate to right shoulder, sometimes even to the neck and upper abdomen.

This may worsen while performing physical activity, usually aggravates on climbing stairs or by performing exercise or by emotional stress or by consumption of heavy meals or by exposure to cold environment and gets relieved by taking either rest or by taking sublingual medicine (sorbitrate 10 mg).

The chest discomfort will be relieved within minutes in case of IHD (see Figure 69) and in case of myocardial infarction (see Figure 70), the pain may be observed for more than 30 minutes.

This may be associated with breathlessness and sweating in the forehead region. If the person is having chest discomfort and it disappears in the patient within seconds, then it should imply that the pain observed in the patient is not of cardiac origin.

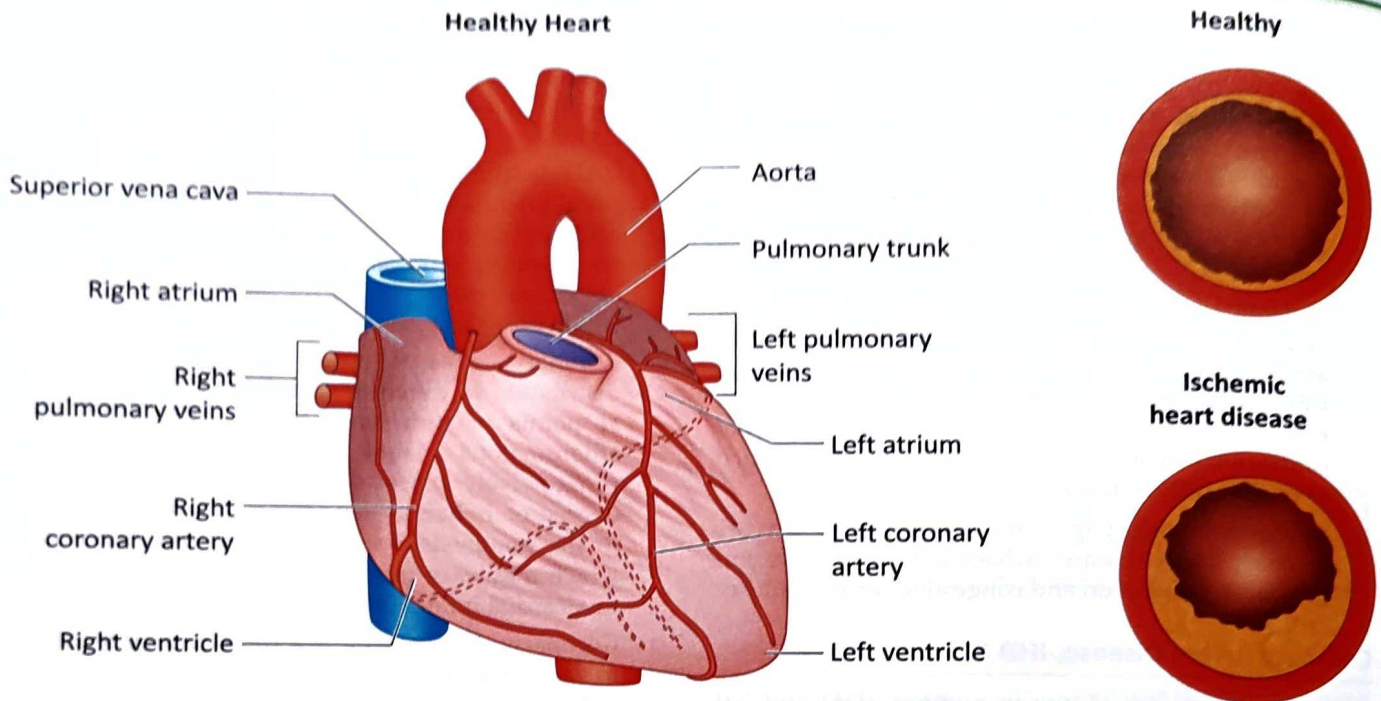


Figure 69: Mechanism of coronary artery disease: cross sectional view of vessel

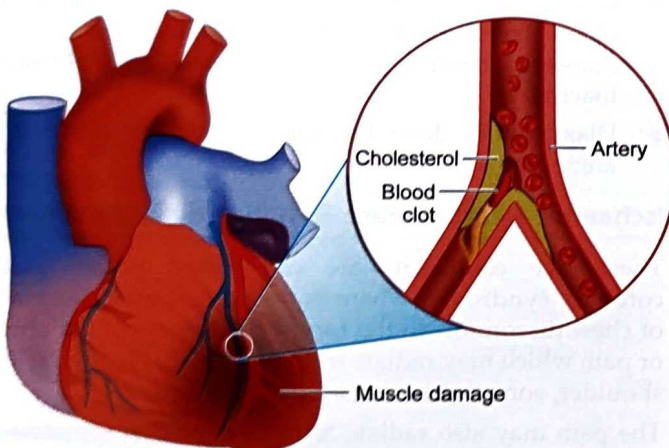


Figure 70: Mechanism of myocardial infarction

- Most common cause of IHD and MI is CAD. The risk factors for the development of these diseases is similar to the risk factors of the development of CAD. Depending upon the duration of the symptoms, IHD is further classified into angina pectoris (where the chest pain gets relieved within 10 minutes. If the person is having pain which persists for few minutes and gets relieved by taking rest or sublingual medicine, then such type of chest discomfort is termed as angina pectoris; and if the pain will persist for more than 10 minutes and less than 30 minutes, such type of chest discomfort is termed as manifested due to myocardial infarction.
- The nature of chest discomfort, location, radiation, character, severity, aggravating and relieving factors

of CAD is observed in the patients of ischemic heart disease and myocardial infarction. Hence physical examination findings of CAD is also similar to the description of IHD, MI.

- Diagnosis is done by CBC, ECG, echocardiogram, TMT, enzymatic assessment of cardiac enzymes.

The differences between IHD and MI:

Ischemic heart disease	Myocardial infarction
Pain observed retrosternal, upper abdominal etc	Pain observed retrosternal, upper abdominal etc
Pain will persist for few minutes (2 to 5 minutes)	Pain will persist for more than 30 minutes
Pain gets relieved by taking rest	Pain will not get relieved by taking rest
Pain gets relieved by taking sublingual medicine (sorbitrate 10 mg)	Pain will not get relieved by taking sublingual medicine (sorbitrate 10 mg)
ECG changes: ST depression and Inverted T wave is diagnostic of IHD	ECG changes: ST elevation (tombstone appearance) and upright T wave

Congestive Cardiac Failure (CCF)

Congestive cardiac failure (see Figure 71) is a syndrome of ventricular dysfunction. Left ventricular failure causes shortness of breath and fatigue etc and right ventricular failure causes peripheral and abdominal fluid accumulation; the ventricles can be involved together or separately. Diagnosis is initially clinical, supported by chest x ray, echocardiography.

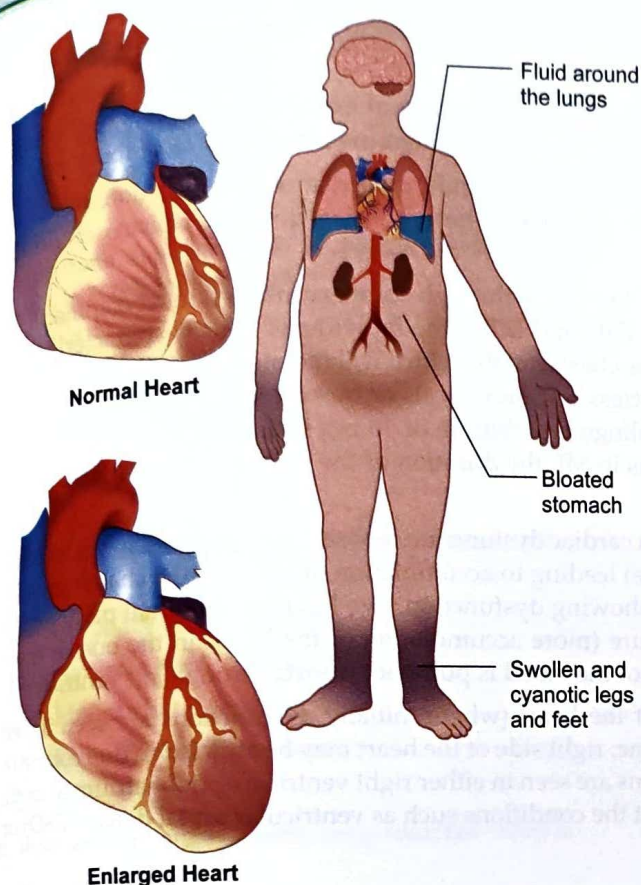


Figure 71: Clinical features of congestive cardiac failure (CCF)

- The term congestion refers to accumulation of some amount of blood in either one or two or three or all four chambers of the heart. Depending upon the nature of pathogenesis, two types of pathogenesis may occur: forward failure and backward failure.
- Aortic stenosis, aortic regurgitation, presence of atheromatous plaques in the ascending aorta are the initiating factors of left ventricular dysfunction. Whereas chronic obstructive pulmonary disease is the initiating factor of right ventricular dysfunction.
- When the person is having poor cardiac output from the left ventricle to the aorta, this may lead to the backflow of the blood to the left ventricle, leading to the congestion of left ventricle. Due to the chronic congestion in the left ventricle, leads to the congestion of left atrium and later congestion of pulmonary capillaries.
- There will be transport of RBCs, fluid etc into the alveoli along with gases, leading to the accumulation of fluid in the alveoli, which is termed as pulmonary oedema. This will lead to the appearance of cough with pink, frothy sputum. (pink due to the presence of RBCs and froth due to the presence of gases in the alveoli).
- When there is backflow of the blood from pulmonary capillaries, excess blood enters into right ventricle via pulmonary artery leading to the congestion of

right ventricle and this in turn leads to the congestion of right atrium and later the blood may move into superior vena cava and inferior vena cava.

- The backflow of the blood in the superior vena cava is identified by analysing the assessment of right jugular vein.
- The backflow of blood in the inferior vena cava can be identifiable by assessment of pain in right upper quadrant and bilateral pedal oedema which occurs as a result of backflow of blood into hepatic veins and inferior vena cava respectively.

When there is backflow of blood leads to the portal hypertension where excess blood move into superior mesenteric vein and splenic vein leading to the backflow of the blood into stomach, small intestine leading to the manifestation of loss of appetite, nausea or vomiting.

- When there is poor forward flow of blood, less amount of blood will move from the left ventricle into aorta. This will lead to poor blood flow into the different body parts of the body.
- Syncope is one such type of symptom which occurs due to poor blood supply to the brain, where the person may have altered state of consciousness which may be observed when the person is upright posture and this may be relieved by making the person to lie in supine posture and loosening the clothes around the neck to facilitate adequate amount of blood supply to the brain.
- Oliguria and nocturia are also observed which occurs due to the poor blood supply to the renal arteries in the day time and increased blood supply to the renal arteries in night time.
- As already stated in the introduction, general and systemic examinations are of great value in the diagnosis of congestive cardiac failure. General examination helps in the detection of visible jugular venous pulse (JVP) on the right side, bilateral pedal oedema specifically seen in evening time, central or peripheral cyanosis in the tongue, fingers, tip of nose, ear pinna (due to circulatory failure, accumulation of deoxygenated blood in the peripheral vessels), clubbing (due to poor circulation).
- Systemic examination helps in the detection of mediastinal shift (shift of apical impulse to the left side); bilateral basal crepitations on auscultation (pulmonary oedema); tender hepatomegaly (due to backflow of blood to hepatic veins); detection of ascites (due to portal hypertension).
- Diagnosis is done by clinical evaluation, chest x ray, ECG, echocardiogram.
- As far as prognosis is concerned, generally patients with heart failure have poor prognosis unless the cause is correctable.

Key summary of the chapter

- ⊗ Many of the symptoms manifested due to cardiovascular dysfunction will show symptoms of other systemic dysfunction.
- ⊗ While investigating cardiac disease, history plays in 40% of the cases, physical examination in 25% of the cases, electrocardiography in 20%, radiography in 10% and special methods of diagnosis in another 5% of the cases.
- ⊗ Symptoms of cardiac disease will manifest because of myocardial ischemia; disturbance in contraction/ relaxation of myocardium; obstruction to the blood flow and arrhythmias. These are 4 major mechanisms of cardiac pathogenesis.
- ⊗ Coronary artery disease (CAD) involves impairment of blood flow through coronary arteries, most commonly by atheromas. CAD is the major cause of manifestation of IHD and MI in the individuals. This is characterized by retrosternal chest discomfort in the form of heaviness of the chest usually, which will radiate to the left shoulder even upto the left fingers; worsens on exercise or emotional stress or heavy meals or by exposure to cold environment and gets relieved by either taking rest or by taking sublingual sorbitrate of 10 mg strength. In case of IHD, the duration of pain will be between 2 to 5 minutes; whereas in MI, the duration of the pain is > 30 minutes and will not respond to rest or by sublingual sorbitrate 10 mg.
- ⊗ Congestive cardiac failure (CCF) is the condition where in cardiac dysfunction is observed in the patient with failure of the chamber of the heart (usually left or right ventricle) leading to accumulation of blood in the chamber of the heart, which is termed as congestion. When ventricle is showing dysfunction, two kinds of abnormal pathological mechanisms can be seen in the body i.e. backward failure (more accumulation of the blood in the one or more chambers of the heart) and forward failure (less amount of the blood is pumped to aorta through left ventricle).
- ⊗ Congestive cardiac failure can also occur in left side of the heart (where initially dysfunctions are seen in left ventricle or left atrium or both, where in due course of time, right side of the heart may be afflicted in due course of time) or right side of the heart (where initially dysfunctions are seen in either right ventricle or right atrium or both, but left side of the heart will be less likely afflicted except the conditions such as ventricular septal defect (VSD) or atrial septal defect (ASD).

Self-Assessment: Questions for Reviews

Short essay questions

- ⊗ Explain coronary artery disease
- ⊗ Describe IHD and MI and write the differences between the two
- ⊗ Explain congestive cardiac failure (CCF)

MCQs

1. Jugular venous pulse isfinding.

- a. Inspection; b. Palpation;
- c. Percussion; d. Auscultation

Answer: (a)

2. Visible jugular venous pulse will indicate the following except

- a. Superior vena caval obstruction;
- b. Right atrial failure;
- c. Right ventricular failure;
- d. None of the above

Answer: (d)

3. The duration of pain in the patient of myocardial infarction observed usually will be

- a. > 15 minutes;
- b. > 5 minutes;
- c. > 10 minutes;

d. > 30 minutes

Answer: (d)

4. Cough with pink frothy sputum is indicative of which of the following?

- a. Right atrial failure;
- b. Right ventricular failure;
- c. Both right atrial and ventricular failure;
- d. Left atrial failure

Answer: (d)

5. Cor pulmonale is the condition which leads to the affliction of

- a. Right atrium;
- b. Right ventricle;
- c. Right atrium and Right ventricle;
- d. Left ventricle

Answer: (c)

6. Which of the following is not an example of forward failure?

- a. Pain in right upper quadrant;
- b. Syncope;
- c. Oliguria;
- d. Nocturia

Answer: (a)

7. Which of the following is an example of forward failure?
- Cough with pink frothy sputum;
 - Nausea;
 - Visible JVP;
 - Syncope

Answer: (d)

8. Tombstone appearance of ST segment with upright T wave in ECG is indicative of
- IHD;
 - Angina pectoris;
 - Acute MI;
 - CCF

Answer: (c)

9. Which of the following is an example of congenital heart disease?
- Coronary artery disease;
 - Ischemic heart disease;
 - Myocardial infarction;
 - Fallot's tetralogy of heart

Answer: (d)

10. In congestive cardiac failure, congestion can occur in
- Left ventricle;
 - Right ventricle;
 - Right atrium or left atrium;
 - Any one or two or three or all chambers

Answer: (d)

Chapter resources

- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Textbook on Clinical biochemistry and haematology with clinical concepts by Naveen Chandra N H, 1st edition, published by the author, 2013.
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- www.longmoreclinic.org for image of coronary artery disease
- www.medexpert.sg for image of IHD
- www.medicoverhospitals.in for image of MI
- <https://my.clevelandclinic.org/health/diseases/17069-heart-failure-understanding-heart-failure>
